

BUILD SPECIFICATION
OF
60 TON BOLLARD PULL ASD TUG

SHIP BUILDER



HINDUSTAN SHIPYARD LIMITED

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SECTION “A” GENERAL**1. PRIMARY FUNCTIONS**

The vessel shall be primarily tasked for ship handling and towing operations within harbour, and in coastal area within 05nm from the nearest land.

The tug shall be able to perform the following operations:

- Berthing/Unberthing
- Escort operations within harbour at speeds less than 6 knots (Ship assist operation)
- Pushing/Pulling
- Fire-fighting

2. AREA OF OPERATION

The ship shall operate within 05nm from the shoreline.

3. OPERATING PROFILE

The ship shall be designed for the following operating profile:

Sl. No.	OPERATIONAL REGIME	% of time of operation
1.	StandBy	35
2.	Transit Low speed (<50% max speed)	25
3.	Transit High speed (>50% max speed)	10
4.	Low Bollard Pull Operation (<50% Max bollard-pull)	20
5.	Medium Bollard Pull Operation (50-90% Max bollard-pull)	8
6.	High Bollard Pull Operation (>90% Max bollard-pull)	2

The exploitation of the vessel shall be approx. 3500 hours/year.

4. SERVICE LIFE

The vessel shall have an expected service life of 20 years.

5. MAIN PARTICULARS

The principal design characteristics of the vessel shall be as follows:

- Length overall abt. 33.0m.
- Length between perpendiculars abt. 31.0m
- Breadth moulded abt. 11.9m
- Depth mid ships abt. 5.4 m
- Hull draught abt. 3.9 m
- Gross tonnage <500 GT
-

6. COMPLEMENT

The vessel shall be arranged with accommodation for a total complement of 11 persons.

7. CAPACITIES

The capacity of fuel oil, lube oil and fresh water shall be designed commensurate with the endurance requirements of the vessel.

Minimum capacities to be provided are as given below:

- Fuel Oil - finalized based on endurance capacity
- Fresh Water - approx.25m3

8. PERFORMANCE

The vessel shall have a static ahead bollard pull of at least 60 tonnes at 100% MCR of the engine.

The vessel shall achieve a speed of approx. **11 knots** at design draught with clean hull at 90% MCR, with wind speed not exceeding BF3.

9. ENDURANCE

The vessel shall be designed for an endurance of 7days with maximum of 18 hours operation per day and operating profile mentioned above.

10. ENVIRONMENTAL CONDITIONS

Machinery shall be able to deliver its specified output and operate satisfactorily under tropical conditions as mentioned below:

- Sea water temperature 32°C
- Air temperature outside 40°C
- Machinery Spaces temperature 50°C
- Relative Humidity abt 60% at 40°C

11. CLASSIFICATION, REGULATIONS, CERTIFICATES

i. CLASSIFICATION

The Tug shall be built in accordance with the rules and regulations of the Indian Register of Shipping (IRS) and according to the following notations:

☞ IWL, TUG, ZONE-1, AGNI-1 (2400 m³/hr)

☞ IY

With Descriptive Note “Indian River-Sea Vessel (Type-1)”

i.e, inland water vessel and machinery built and installed under supervision of the Surveyor of IRS to the “Rules and Regulations for the Construction and Classification of Inland waterways Vessels-2024”. The construction shall be as per standard shipbuilding practice with good quality materials. The vessel shall sail under Indian Flag.

ii. RULES AND REGULATIONS

Tug shall be in compliance with all relevant International Standards as applicable at the time of construction, including (but not limited to) the following:

- a) COLREGS-International Regulations for Preventing Collisions at Sea
- b) Load Lines, 1966/1988- International Convention on Load Lines, 1966, as Amended by the Protocol of 1988
- c) MARPOL –International Convention for the Prevention of Pollution from Ships
- d) Tonnage-International Convention on Tonnage Measurement of Ships, 1969
- e) 2008 IS Code–International Code on Intact Stability, 2008
- f) International Labour Conference– Maritime Labour Convention, 2006
- g) IMO regulations A468 (Noise levels) and ILO Recommendation R141 as far as practicable for tugs.
- h) International Convention on the Control of Harmful Anti-Fouling Systems on Ships, 2001

iii. CERTIFICATES

The Builder shall obtain following certificates/equivalent and deliver to the Owner at the delivery of the vessel.

- a) Certificate of Classification for Hull and Machinery Survey

- b) Bollard Pull Certificate
- c) International Tonnage Certificate
- d) Builder's Certificate.
- e) Trim and Stability Booklet.
- f) All other certificates as required by Class.
 - Certificate of Class
 - Indian River Sea Safety Certificate
 - AFS CERTIFICATE
 - MLC

12. BUILD METHOD AND WORKMAN SHIP

Generally, all works shall comply with the requirements of the Classification Society and other Authorities governing this specification.

Steel work to be carried out in compliance with IACS Rec.047 Part A "Ship building and Remedial Quality Standard for New Construction".

All works shall be carried out in accordance with the approved drawings.

The electrical installation to be according to IEC norms.

All equipment and machinery to be installed according to manufacturer's instructions.

13. DRAWINGS& DOCUMENTS

Required documentation to Regulatory Authorities, Classification Society and Owner shall be handled by the Builder.

All documentation shall be in English language.

Method of drawing including the scales to suit Builder's standard practice.

14. OWNER APPROVAL

Builder shall submit, to the Owner a set of Classification Documents for their approval. Owner and Builder shall decide the method of submission and schedule for the approval of the drawings separately be for commencing the project.

15. CLASSIFICATION DOCUMENTS

Builder shall prepare and submit for approval to Classification Society all documents that mandate Class approvals. Builder is obligated to incorporate all comments and remarks made by Classification Society.

16. DOCUMENTS TO BE PROVIDED AT DELIVERY OF THE VESSEL

The Builder shall handover at least the following drawings and manuals to the Owner at the time of delivery of the vessel.

- a) General Arrangement Plan
- b) Final Stability Booklet
- c) Mid ship Section
- d) Transverse Sections including bulkheads
- e) Shell Expansion Plan
- f) Profiles & Deck Plan
- g) Super structures/Deck Houses
- h) Skeg Structure
- i) Docking Plan
- j) Machinery and Propulsion Arrangement
- k) Piping diagram schematic
- l) Navigation Lights and Sound Signals Plan
- m) Life Saving Appliances Plan
- n) Fire Fighting Plan
- o) Electrical Single Line Diagram
- p) Electrical Wiring Diagram
- q) Electrical Equipment Arrangement
- r) General Service and Radio Service battery load Analysis
- s) Electrical Load Analysis
- t) Bridge Arrangement
- u) Accommodation Plan
- v) Sounding Tables
- w) Towing Arrangement
- x) List of Machinery, Equipment and Plants of Third Parties' Supply, with following Details: name of manufacturer, address, and telephone no.
- y) Capacity Plan
- z) Hydrostatic Curves

In addition, any drawing that needs to be displayed on the vessel as per Class or Regulatory Authority guidelines shall be properly framed and placed onboard the vessel by the Builder.

17. INSTRUCTION BOOKS

Instruction books of the vessel's major machinery, i.e., Main Engines, Azimuth Thrusters, Generators and other major equipment and systems shall be furnished to the Owner. (The number of copies that needs to be furnished shall be finalized by the Owner and the Builder at the time of signing the Ship Building Contract)

The builder shall also prepare data booklets for Hull, Machinery and electric equipment giving the name/address of makers and references to such information as maintenance instruction books and/or spare parts list.

18. EQUIPMENT AND SPARES

Spare parts shall be supplied by the Builder according to the OEM Specifications, which are mandated by Classification Society and Regulatory Authority for the intended operation.

SECTION “B” HULL

1. HULL MATERIALS

STRUCTURAL STEEL

(a). The Hull shall be built of entirely of shipbuilding quality Grade “A” steel and steel for other requirements (other than main hull) to be of IS-2062. The scantlings of main Deck, Shell and Transverse WT Bulkheads should be 10% excess of rule requirements.

(b). The fore and aft sections to be strengthened for pushing operations.

(c). The vessel shall be of welded construction throughout. Welding is to be carried out in accordance with Classification requirement. Surveyors shall approve schedule of welding and plate edge preparation. Welding in important places like keel, Stem, Skeg, Propeller Guards, Propelling Unit Seating's, Engine Foundations etc. should be subjects to X-Ray test after welding.

(d). The Hull structure should be so designed that there should not be any abnormal vibrations of the structure in the operating range of speeds.

The materials used shall be of good international shipbuilding standards. Steel plates, forgings and castings shall be delivered with certificate according to the Classification Societies and Regulatory Authorities' requirements. Test marks and makers name shall be clearly stamped there on.

2. TESTING OF TANKS, BULKHEADS

Air pressure tests shall be done for all tanks as required by Classification rules.

Hose testing of doors, hatches and other shell side and deck openings shall be done according to Classification Society requirements. Hose testing shall be carried out on all watertight structures, which are not tested by water or air pressure.

STRUCTURAL TESTING

Structural testing (hydro-test) for tanks (except voids) shall be done as per class requirements.

The tests shall be performed in the presence of Class Surveyor and Owner Representative. Owner shall be intimated about the test schedule in advance. Test reports shall be submitted to the Owner.

3. NON-DESTRUCTIVE TESTING (NDT)

NDT shall be carried out as per Class requirements.

X-ray testing shall be carried out as per Classification Society requirements and class approved NDT Plan.

Ultrasonic or dye-penetration test of weld shall be carried out to the satisfaction of Classification Society where it is not possible to undertake X-ray testing.

4. WELDING

Welding shall be carried out in accordance with the requirements of classification society.

Welders qualified through standard welding procedures approved by Class shall carry out welding of hull and all main structural elements.

5. AFT BODY

The thickness of the shell plating shall be in accordance with class requirements. Increased thickness may be considered if required i.w.o. thruster units, based on structural analysis and recommendations from supplier, if any.

Aft body shall be suitably stiffened to prevent vibration. The thrusters shall be supported by deep solid floors.

A double plate Skeg of adequate size shall be fitted on the centre line. Skeg depth shall be finalized based on the navigational draught restrictions and dimensions of the selected thruster.

6. ENGINE & MIDSHIP AREA

Engine room area (especially i.w.o of engine foundations) shall be adequately stiffened to minimize the vibration effects.

7. ENGINE FOUNDATION

Adequate strengthening shall be ensured for structure i.r.o. of engine foundations.

The design of foundation shall be as per class requirement and in accordance with manufacturer's recommendations.

The design shall ensure good connection of the foundation with the

supporting/ connecting structure for load transfer.

8. BULKHEADS AND TANKS BELOW MAIN DECK

BULKHEADS

The watertight Bulkheads as shown in the general arrangement drawing are to be placed horizontally. Vertical stiffeners are to be spaced at 500mm apart. Where pipes etc. are passed through the watertight bulkhead are to be provided with watertight arrangement to the satisfaction of Classification surveyors.

Bulkheads shall be suitably stiffened in accordance with Class requirements.

HULLTANKS

Hull integrated tanks shall be arranged according to the general arrangement drawing.

Separate Service/Day tanks shall be provided for engine room as indicated in the GA drawing. The capacity of the day tanks shall be decided as per class/statutory requirement.

9. CARGOAREA

Deck loading in the aft working deck shall be 5 t/m² or as agreed between the builder and the Owner. The vessel shall have a working deck area of at least 65m².

10. STEM

To be of rolled plate type with heavily welded steel frame swept into keel sections at foot and flat above water line for bow fender and fitted with suitable breast hook and stiffeners to be arranged for maximum strength.

Increase in plating thickness may be considered, if necessary, at near vicinity of anchor handling or where large local loads are expected, but need not be greater than class rule requirements.

Collision bulkhead shall be provided in accordance with class rules.

11. FRAMES

Frames are to be extended from keel to deck level and spaced not more than 500mm throughout. The frames are to be welded to shell plating. Strong web frames are to be fitted in the engine room, in way of deck transverse with particular attention being paid to the area adjacent to the propulsion units.

12. FLOORS

Solid floors are to be fitted on every frame throughout with flat bars welded to the floors. Drain and limber holes are to be provided to drain all water to pump suction.

13. CHAIN LOCKERS

Chain lockers to be located suitably with 2 compartments side by side for port and starboard anchor cables. Size of chain lockers shall be sufficient to accommodate full length of the chain. The cable ends shall be anchored with suitable arrangement and drainage of cable lockers to be provided as per classification rules.

14. DECK HOUSE AND SUPERSTRUCTURE

The deck housing superstructures, masts etc. should be so structured and fitted as to allow the tug, to work without being damaged when working in close contact with the tow and with heeling angles up to 20°.

The superstructure shall be arranged in accordance with the GA drawing approved by Owner. The structural scantling shall be in accordance with class requirements.

15. CAMBER

Deck should have adequate camber so that water will drain off easily.

16. MAST

A steel mast for navigation lights and antennae is to be located forward of the funnel on the bridge top deck at centerline and integrally supported.

Mast height and mast fittings including Navigation lights signal flag yard, radar, antennae etc. shall be determined in accordance with relevant Class/Statutory Rules.

Masts to have rungs carried on top, arranged for access to light trays and necessary fittings. Safety cage to be fitted if necessary.

17. BRIDGE

The bridge shall be designed and built with as near as possible 360° unobstructed visibility.

Bridge shall be provided with clear, wide and large size windows

wherein each bridge window shall be fitted with wipers. The funnel aft of the bridge shall be so arranged that the funnels do not obstruct the aft view from the engine controls in bridge. Suitable arrangement shall be provided for washing Bridge front windows.

The Bridge shall be fitted with three window wipers one on forward and one each on the Port and Starboard side ends. The windows shall be with proper Aluminium alloy frames and toughened glass of not less than 8mm thick.

As many number of windows as possible shall be capable of being opened so that glare can be minimized during night time. Provision shall be made for cleaning whenever required and to drain off water at the sills.

Tinted sky windows are installed slantwise in the Bridge top deck fitted in metal frames.

All steering control equipment shall be located at Bridge.

18. HULL OUTFITTING

HULL MARKINGS

All markings shall be as per best global standards.

The ship's name shall be marked with suitable materials and mounted on the superstructure port and starboard (P&S).

Other hull markings like Port of Registry, IMO number, Owner's logo etc. shall be provided at appropriate locations, as agreed between the Builder and the Owner, as per global best shipbuilding practice.

Draught marks shall be marked using weld beads/ steel plates and shall be painted in white on bow and stern on P&S sides. Horizontal marks at every 100 mm distance, metrical scale shall be followed.

The vertical extent of the draught marks shall be adequate to cover the draught of the vessel under all probable conditions of loading and corresponding trims with the vessel undamaged.

All machinery equipment, valves, hand wheels, levers, doors, ventilation etc. shall be indicated by nameplates of plastic/ suitable non-corrosive metal with black letters in English.

Bottom plugs and manhole covers shall be marked with tank number and content using weld beads. Watertight bulkheads, tank boundaries, number and contents shall be marked on hull and deck.

Loaded water line shall be marked using appropriate paint as per global best shipbuilding practice.

19. BOTTOM PLUGS, BILGE WELLS ETC.

One (1) bottom plug each shall be supplied and fitted in hull tank, cofferdam, etc. where considered necessary.

The size and design shall be as per global best ship building practices.

Water tanks shall have square and oil tanks shall have hexagonal sockets respectively.

Special attention shall be given to locate the bottom plugs as close as possible to the lowest point of each tank. The bottom plugs shall be kept clear of points designated for keel blocks as per docking plan.

Bilge wells and sea tubes shall be arranged as necessary at suitable places in the double bottom.

20. FOUNDATIONS

All auxiliary, deck machinery, electrical equipment etc. shall be erected on foundations.

As far as practicable, all deck machinery foundation shall be open type with accessibility for cleaning and painting works. Suitable reinforcements and insert plating shall be provided under the machinery, wherever necessary. Proper draining arrangement shall be provided to avoid accumulation of water.

21. FENDER, BILGE KEEL

i. Fender

Fendering will be provided with “W/D – type fender” all around the vessel, with additional tyre fenders. Tyre fenders are to be fitted around the vessel such that 4 nos. large tractor/truck/earth moving equipment tyres are arranged at the bow and heavy-duty air craft tyres abutting either side of the vessel.

ii. BilgeKeel

Bilge keels are to be provided on the vessel.

Welding of bilge keel is to be performed as per Class approved design drawings. Preferably, notch welding shall be applied iwo bilge keels and the bilge keel has to be welded on doubler plate against

shell plating.

22. HAWSE PIPES

The hawse pipe shall be fitted in such away that free anchor fall is obtained.

The hawse pipe shall have adequate diameter and length to easily house the anchor shank and shackle.

Chafing ring of steel bars shall be fitted at the lower edge of chain pipe for protection and smooth running of chains.

23. BULWARKS

Strong bulwarks of steel plate, suitably stiffened shall be fitted all round the vessel at main deck and forecastle deck.

The bulwark shall have tumblehome and access doors shall be provided on both port and starboardside in the main deck level.

The height of the bulwark shall be 1.0m and with stays spaced not more than 1.2m apart.

The top of the bulwark shall be reinforced with a flat bar /steel pipe of suitable dimension/ thickness.

24. MATERIAL PROTECTION

a. PAINT SPECIFICATION

The paint system shall be of good quality marine grade. The final paint specification shall be mutually agreed between Builder and Owner, as per the recommendations from the paint maker.

Final colour to be confirmed by Owner.

Paint in under water hull, water ballast tanks, and fresh water tanks, to be guaranteed for five (5) years. All dry film thicknesses shall be as recommended by supplier for a five (5) year system.

Paint specification and surface preparation to be of high standard, where in prior to fabrication, all steel work to be shot-blasted to SA 2.5 and shop-primed. The surface preparation and paint specification shall be in compliance with SOLAS with respect to toxic and environmental requirements. Surface treatment shall be given to all welds, sharp edges, undercuts, slag, etc.

Decks to be coated with high friction anti-skid paints, in alignment with the paint scheme.

All edges, corners, logs, weld seams etc. to be stripe coated with

brush or roller, between each layer according to paint maker's instructions.

Descaling, shop priming, de-rusting and painting works shall be carried out in accordance with yard practice.

All necessary remedial works would be carried out to owners satisfaction, based on the observations during inspections, before delivery.

b. Paint Scheme

Painting scheme in general shall be as given below. Detailed paint scheme covering all areas of the vessel shall be prepared in consultation with paint manufacturer for a 5 -year system.

LOCATION	PAINT SCHEME	DRY FILM THICKNESS
Under water Hull	2 coats of tar free epoxy 1 coat of sealer 2 coats of tin free anti-fouling	DFT as per paint maker recommendation for 5-year life
Top sides	3 coats of Re-coatable epoxy	DFT as per paint Maker recommendation for 5-year life
Inside machinery spaces/ Accommodation/ Thruster Room/stores etc.	2 coats of alkyd paints/as agreed between builder and Owner	DFT as per paint Maker recommendation
Ballast Water & FW Tanks	As per paint maker Recommendation for 5-Year life	As per paint maker Recommendation for 5-Year life
Main deck/Raised Main Deck	As agreed between builder and Owner	As per paint maker recommendation
Super structure	As agreed between builder and Owner	As per paint maker recommendation

		on
Funnel	As agreed between builder and Owner	As per paint maker recommendation
Void spaces	As agreed between builder and Owner	As per paint maker recommendation

25. EXTERNAL CATHODIC PROTECTION

i. Sacrificial Anodes

Ship's external hull shall have cathodic protection against corrosion for 5 years. Anodes with high degree of purity shall be used where applicable.

26. SMALL HATCHES & MANHOLES

i. Manholes

Bolted manholes shall be provided, in number and location as necessary, as per class requirement for access to all compartments, tanks, cofferdams, voids and pockets not provided with other means of access.

ii. Hatches

Single lever type watertight access hatches with steel covers shall be provided on weather deck.

For the removal of main machinery during overhaul/ maintenance, bolted hatches are to be installed on the main deck on top of Engine room.

SECTION “C” OUTFIT & ACCOMMODATION

1. ANCHORING, MOORING & TOWING EQUIPMENT

Anchoring, mooring equipment and towing arrangement shall be provided as per Classification Society requirements.

2. ANCHORS WITH CHAIN & EQUIPMENT

Anchor and Chain cables: Two HHP stockless type anchors of adequate weight as required under classification rules for this size of vessel and stowed with suitable arrangements on outer shell plating with required shackle lengths, on Starboard and port side of required diameter, Grade U2, stud link cables in 15 fathoms length joined with kenter shackles. No anchor pockets will be provided for anchor stowage on shell.

Windlass: An anchor windlass of adequate capacity capable of heaving up anchors as per classification rules shall be mounted on fabricated steel seating on forward main deck. Cable lifters shall be of Cast steel with gunmetal bushes and properly shaped to give full grip to anchor cables and independently clutched with brakes. Two warping barrels shall be provided. Hawse pipe shall be correctly positioned to give cables an easy lead to windlass.

3. MOORING EQUIPMENT:

Mooring equipment shall be provided as per class requirement.

Mooring rope, towing rope, heaving line, hawser lines as per applicable class rules shall be provided.

One (1) off Electric Mooring capstan with line pull & controls near the capstan.

Bollards, cruise form bollards & bitts shall be as shown in General Arrangement. Arrangement of mooring equipment to be finalized with owners.

4. TOWING HOOK

A spring loaded towing hook of 60 tons safe working load and capable of horizontal movement up to 35° on either side, shall be provided. The tow hook shall be provided with quick releasing system, which can be operated from the Bridge and from the main deck near the hook. Special attention shall be taken for stiffening of the structure in the region of towing hook.

Sufficient rope guiding arrangements such as tow pin/ gob eye/A-frame may be provided on the aft deck, based on requirement from the Owner.

5. LIFESAVING EQUIPMENT

a. RESCUE BOAT

One (1) no. rescue boat of suitable type that can be lowered and hoisted manually on to the vessel by 2 persons shall be provided meeting the requirements. Mechanical lowering arrangement for lowering and hoisting by electric motor of rescue boat shall be provided.

b. LIFERAFTS

One or more inflatable life rafts (SOLAS Pack B) with relevant certification shall be provided. They shall be mounted on either side of bridge deck with suitable supports.

All life rafts shall be in accordance with the latest SOLAS requirement and shall be provided with hydrostatic release arrangement.

The life rafts shall be stowed in self-launching cradles and secured with release lashing.

c. LIFE SAVING EQUIPMENT

Required life buoys, life jackets and other LSA items shall be provided to meet the class rule requirements.

6. FIRE FIGHTING EQUIPMENT

Fire fighting equipment shall be provided on the vessel in compliance with Class requirements as applicable at the time of vessel construction.

FRP cabinet shall be provided for securing fire hoses and nozzles.

Sufficient number of fire man's outfit, breathing air compressor, etc. in compliance with AGN11 or equivalent notations to be provided on the vessel.

7. ACCOMMODATION–INSULATION, PANELS, DOORS, WINDOWS, LOCKERS

7.1 General

Class approved materials shall be used for the vessel as applicable, including fire retardant paints, curtains, linings, etc.

7.2 ACCOMMODATION

Accommodation shall be provided for 2 Officers and 9 Crew members

For all compartments, required minimum clear height shall be ensured.

The vessel shall be arranged with the following facilities in cabins:

Master/Chief Engineer Cabin

- CRCA berth with drawers
- CRCA Wardrobe with mirror
- Writing table with locker
- Up holstered arm chair
- One settee
- Document Locker/ Shelf
- Coat hooks
- Glass & tumbler holder
- Book rack
- Wall mounted fan
- Attached Toilet

Other Cabins

- CRCA berth with drawers
- CRCA Wardrobe with mirror
- Writing table with locker
- Arm chair
- Coat hooks
- Glass & tumbler holder
- Book rack
- Wall mounted fans

7.3 PARTITION BULKHEADS, PANELLING

In general, thickness of linings shall be 25 mm and that of partitions shall be 50 mm. The partitions shall be of non-combustible panels having good quality surface finish.

Low flame spread sandwich panels shall be provided for ceilings.

0.9mm Stainless steel Ceiling and partition panels shall be provided in galleys.

8. DOORS

In general, accommodation spaces shall be fitted with joiner doors and fire rated doors as applicable. Weather tight doors are to be provided for exterior access to Bridge and lower deckhouse

Water tight doors shall be provided on the main water tight bulkheads. All water tight doors are to be fitted with hinges and toggles in compliance with class requirements.

The Bridge doors shall be hinged and provided with glasses for good visibility.

Sill height to be in accordance with the relevant Rules and Regulations.

9. WINDOWS

All accommodation areas are to be provided with windows/ scuttles/ skylights as per the requirements of Regulatory Authorities and Classification Societies.

All windows are to be made of hard glass and fitted in metal frames.

10. INSULATION

Insulation for fire, heat & sound shall be provided in accordance with class/ statutory requirements, conforming to global best practices in tug construction.

11. DECK COVERING, LADDERS, STEPS, RAILINGS, GANGWAY ETC.

In general, floorings, ladders and other fittings shall be provided as described below:

- a) Wet deck covering shall be provided in wet spaces like galley and toilet spaces.
- b) Suitable flooring arrangement i.e Steel chequered floors, shall be provided in machinery spaces as agreed between Builder and Owner.
- c) Suitable deck composition (dry deck covering) shall be provided in chief officer's cabin, bridge and dining space.
- d) Ladders, stairs and handrails shall be arranged in accordance with the General Arrangement Plan and where required. If it is impracticable to have either inclined or vertical ladder, the steps of square bar shall be directly fitted on the surface.
- e) Galvanized steel handrails of suitable diameter shall be welded along the sides and front of the deck house.

12. BRIDGE

The Bridge shall be arranged with all necessary equipment, as prescribed by the Classification and Statutory Regulations, which includes but not limited to the below.

- Consoles
- Chart table
- Flag locker
- Binocular holder
- Revolving Arm chair
- GMDSS Table

Special attention shall be given to the overall ergonomics while arranging Bridge items.

13. GALLEY/PANTRY EQUIPMENT, PROVISION PLANTS, LAUNDRY/ IRONING EQUIPMENT

13.1 GALLEY

Galley shall be provided with the following equipment:

- a) Electric marine cooking range with oven
- b) Full size refrigerator with freezer
- c) Stainless steel side board
- d) Water Boiler
- e) Stainless Steel Sink with fresh water supply
- f) Stainless steel work table
- g) Canopy with electrically-driven exhaust fans

13.2 PROVISION STORE AND REFRIGERATION SYSTEMS

A provision store shall be provided in the main deck.

Deep freezers of adequate capacity shall be installed in provision store for carriage of meat, fish, vegetables, dairy products, etc.

14. VENTILATION, AIR CONDITIONING & HEATING SYSTEMS

14.1 General

The air conditioning plant shall be designed for the following parameters:

External Temperature	: Maximum 41°C (RH-90%)
Internal Temperatures	: 27°C (R.H 50%).

In all air-conditioned compartments (except Galley)

Design temperatures may be reduced suitably based on area of operation.

14.2 VENTILATION/AIR-CONDITIONING SYSTEMS FOR ACCOMMODATION, CONTROL SPACES ETC.

Entire accommodation area, Bridge and switch board room shall be air conditioned.

Centralized Air-conditioning unit with fan-coil and fresh air handlers etc. will be provided.

14.3 VENTILATION SYSTEM FOR MACHINERY SPACE

The machinery space shall be ventilated by a mechanical ventilating system.

The ventilating fans shall supply fresh air to the operating station and other necessary places in the engine room through air ducts. One off supply fan in engine room shall be of reversible type.

The exhaust air from the engine room shall be led to the atmosphere through openings provided on the funnel.

The exhaust air opening and fresh air intakes for engine room shall be provided with emergency shut-off fire dampers operated from the outside of the engine room.

Mechanical ventilation shall be provided for service/ machinery spaces including thruster room, battery room, workshop etc.

In addition, galleys and toilets shall also be provided with suitable arrangements for forced ventilation.

15. DRINKING AND SANITARY SERVICE SYSTEM FOR ACCOMMODATION

15.1 Fresh Water Service System

The system shall supply fresh water for sanitary services in accommodation, equipment in machinery spaces, galley and laundry.

15.2 SHOWERS AND TOILETS

Toilet spaces with showers shall be provided as shown in the general arrangement plan. Toilets shall have gravity type discharge.

15.3 SANITARY DISCHARGE SYSTEM

In general, sanitary discharge system shall consist of sewage-holding tank and shore discharge pump.
Shore Discharge flange to conform to MARPOL 73/78, Annexure4, Regulation 10.

SECTION "D" MACHINERY

1. MACHINERY- GENERAL

The machinery specification consists of the main particulars of the propelling engines, propulsion units, auxiliaries etc and Shipyard shall ensure to provide all equipment and machinery required for the total efficient performance of the Tug. The main engine and auxiliary machinery shall be designed to withstand static and dynamic inclinations in accordance with IRS Rules. All accessories and fittings for Machinery shall be as per respective maker's standard complying Rule requirements.

2. PROPELLING UNITS

(a). Two (2) numbers fixed pitch steerable rudder propulsion units of suitable capacity shall be installed on Port and Starboard sides, each one mainly consisting of a steerable rudder propeller drive by a four stroke marine diesel engine via a universal type intermediate shaft. The units shall be remote controlled from Bridge.

(b). The unit shall be provided with hydraulic powered steering. The hydraulic installation for powering the hydraulic operated equipment shall consist of a pump driven by a motor. The units along with hydraulic oil tanks of sufficient capacity shall be provided in thruster room.

(c). All recommendations of the propulsion unit suppliers on the Hull construction shall be strictly followed.

(d). Duplex control stand installation for machinery control transmission from bridge to propeller units consisting of 2 control stands, each with hand wheel and two speed levers including machinery connecting lines between control stands for synchronization as well as mechanical control to steerable rudder propulsion shall be provided. Control stands shall be as per maker's standard.

(e). The main engine and SRP units shall be started from engine room and then control shall be transferred from engine room to bridge and vice versa while stopping the propelling machinery.

(f). Emergency steering of the vessel shall be as per statutory/class rules. Steering system shall comply with the class requirements.

3. PROPELLING ENGINES

(a). Two (02) numbers four stroke, turbocharged, non-reversible, unidirectional, intercooled, direct injection, pneumatic started, propulsion marine diesel engines shall be provided to match the propelling units for

meeting the requirement of minimum 60 tonnes bollard pull. The vessel is designed to operate, shall be installed on the vessel to meet specified endurance and speed. Engines shall be resiliently mounted if required, to reduce vibration.

(b). The grade of fuel to be used in the engine shall be Low Sulphur High Flash High Speed Diesel (LSHFHSD) oil.

(c). The Main Engines shall be started by compressed air and equipped with all necessary fittings and accessories, such as Duplex L.O. filter, Duplex Fuel oil filter, air starting valves and air filter, fuel pump, lubricating oil cooler, fresh water cooler, Lub oil pump, F.W. pump, exhaust manifold, silencer, Pressure/Temperature gauge panel, cylinder relief valves, tachometer in the engine room and remote tachometers in the bridge. Silencers are to be provided with spark arrestors.

(d). The critical speeds of main engine should be avoided as far as practicable, within range of normal operation speeds.

(e). One complete set of tools for maintenance of the engines including crank case deflection gauge shall be provided with the engine. Each engine shall be assembled at makers works for tests in the presence of the classification surveyor. IRS recommended spares to be provided.

(f). All coolers shall be of plate type heat exchangers.

(g). Provision for PTO for external Fi-Fi pump to be provided on the engines, depending on the configuration of External Fire Fighting Pumps

(h) **MAIN ENGINE ALARM SYSTEM:**

Alarm panel shall be fitted in the engine room in suitable position. This alarm panel is to consist of audible and visible signals for the following;

1. High Cooling Water temperature
2. Low Lub Oil Pressure
3. Over Speed.

Other alarms including shut downs shall be provided as per IRS rules. Fuel cut off shall also be provided for too high cooling water temperature, too low L.O. pressure and over speed as per IRS rules.

(i). **ENGINE EXHAUST PIPES AND SILENCERS.**

Each engine exhaust to pass through a length of flexible piping/bellow immediately after the Engine and through M.S. Pipes in the funnel all designed to withstand expansion and vibration. Inspection at the uptake and condensate drain with stop valves shall be provided.

Exhaust pipes to have an easy lead from silencers to the outlet and lagged with ceramic fibre insulation blanket to withstand the temperature and covered with fibre –reinforced plastic (FRP) cloth or galvanized sheet iron.

Silencers and spark arrestors of main and auxiliary engines shall be well secured in the funnel. Each silencer to be provided with hand holes for inspection as per maker's standard.

- (j).Lifting arrangements with shackle shall be provided over engines for maintenance works, the overhead beam and lifting trolleys and chain blocks. Capacity shall be to lift largest engine part for normal maintenance of main engine and SRP unit.
- (k). Power transmission through coupling or otherwise to suit the propulsion system offered shall be provided.

4. SHAFTING AND MAIN REDUCTION GEAR

Shafting system shall have sufficient strength as per the class rules to take maximum engine power when undertaking high bollard pull operations. The material of shaft shall conform to internationally accepted norms.

Cardan shafting i.w.o. bends, if required shall be of sufficient strength, meeting class requirements.

Adequate arrangement shall be provided for covering the shafts using chequered plate as per global best practices in tug construction.

5. AUXILIARIES

- (a). 2 No's (1 working + 1 standby) sea water cooling pumps , each of capacity to cater for 2 main engines + 2 SRP units.
- (b).Three (03) independent motor driven Main air compressors of approx. capacity 30 Cu.m/hr at 30 Kg/ Sq. cm each shall be connected to air bottles with a capacity of 500 litres each to give 12 starts of each engine without replenishing. Main Air Compressor shall start and stop automatically depending on pressure in Air bottle. Necessary Pressure switches, Pressure Gauges, Relief valves, etc shall be provided.
- (c). Two number (1 working + 1 standby) Fuel oil transfer pump of 8 Cu.m/hr at 25 meters capacity each shall be provided for transfer of fuel oil from one storage tank to .another storage tank.
- (d). 2 No's (1 working + 1 standby) domestic fresh water hydrophore pump

of 2.5 Cu.m/hr at 30 MWC and 500 litres capacity hydrophore tank.

- (e). One number Bilge pump of 35 Cu.m/hr at 3 bar capacity vertical centrifugal electric driven type shall be provided. For pumping out bilge from compartments other than engine rooms to overboard, a suitable capacity hand pump with piping arrangement to be provided
- (f). One number Bilge/GS/Fire pump of 35 Cu.m/hr at 3 bar capacity vertical centrifugal electric driven type shall be provided. One number Fire pump of 35 Cu.m/hr at 3 bar capacity vertical centrifugal electric driven type shall be provided. Both Fire and Bilge/GS/Fire pumps shall cater for onboard firefighting and deck wash system.
- (g). Bilge main system to be complete with suction from all watertight compartments. Engine Room shall be provided with two nos direct bilge suctions. Bilge alarms are to be installed in each watertight compartment.
- (h). An independently engine driven emergency fire pump (Portable) having a minimum capacity of 25Cu.m/hr shall be provided for emergency purpose.
- (i). Dedicated Oily water and sludge tanks of sufficient capacity shall be provided. A motor driven oily water pump and a screw type sludge pump shall be provided for pumping out oily water and sludge respectively.
- (j). All the pumps should be installed in a position such that they have positive sea water suction.

6. DIESEL GENERATOR SETS.

- (a). Two nos Diesel Engines, 4- stroke, compressed air/ battery starting coupled to meet the alternator capacity of 140KVA shall be installed complete with all equipment and accessories as per maker's standard.
- (b). The generator sets shall be placed with their axis of rotation in the force of fore and aft direction. The diesel engines/alternators shall be of indigenous make.
- (c). One number indigenous make harbor generator of approx.. 65 KVA shall also be provided with maker's standard accessories and fittings. Final rating shall be as per electrical load calculation during detailed design. Harbour DG set shall be started by batteries.

7. EXTERNAL FIREFIGHTING SYSTEM

An external fire-fighting system shall be fitted, comprising of engine driven PTO pumps, fire monitors and water spray system, meeting AGN1 or equivalent notation.

Fire pumps may be arranged in any of the configurations below:

Configuration 1:

One no. fire pump connected to main engine PTO. During FiFi operation this engine will be declutched from thruster using On/Off clutch and vessel position is maintained using the other engine and thruster. The fire pump shall have a dedicated sea chest.

Configuration 2:

Two no. fire pumps connected to main engine PTO. Thrusters (in case of FPP) connected to engine through integrated medium duty slipping clutches. Each fire pump shall have dedicated sea chests which are interconnected.

Any other configuration may also be considered as mutually agreed between the Owner and the Builder.

Fi-Fi Pumps

The Fi-Fi pumps shall provide sufficient water to two fire monitors and water spray system.

- Type : Centrifugal
- Drive : Engine PTO
- Total Capacity : min. 2400 m³/hr

(meeting AGN1 or equivalent notation)

Foam System

Foam tank(s) of sufficient capacity complete with necessary fittings shall be provided. The system shall be provided with line proportioner for foam mixing. The foam type shall be AFFF.

Fi-Fi Monitors

Two (2) single barrel dual purpose monitor search having capacity as per AGN1 notation shall be mounted on bridge deck as indicated in the General Arrangement drawing.

- Capacity: Two nos each of min. 1200 m³/hr will be provided

meeting the notation requirements (AGNI I or equivalent notation)

- Min. height of throw : 45m
- Min. length of throw : 120m

The fire monitors may be mounted either on the bridge deck (as shown in the GA) or on Bridge top.

Water Spray System

The vessel shall be provided with self-protection measures as required by AGNI -I or equivalent notation.

These measures shall include fixed water-spraying system for protection of all outside vertical area of hull, superstructures and deck houses including foundations for water monitors and other equipment.

The capacity of fixed water-spraying system shall be provided to meet class requirements. The capacity shall not be less than 10 litres per min. per [m²] of the areas to be protected. For areas internally insulated to class A-60, however, a capacity of 5 litres per min. per [m²] may be accepted.

The arrangement for water-spraying system is to be such that necessary visibility from the Bridge and the control station for remote control of the fire fighting water monitors can be maintained during water spraying.

Fi-Fi Control System

Fire monitors shall have controls for both bridge and local operations. Where in, a fixed control panel shall be mounted in the bridge with the operator having unrestricted view on the Fi-Fi monitors.

8. SHIP'S FIREFIGHTING SYSTEM

(a). Ships internal firefighting arrangements shall be provided to rule requirements for this size of vessel and connected to general service pump.

(b). Fire Hydrants shall be located at all important places and connected to G.S/Fire Pump.

(c). The ships firefighting system shall also include extinguishers for specific firefighting system for dealing with oil and electric fires.

(d). Fire hydrants of Single delivery with 63mm with instantaneous coupling. 2 Nos on Port & Starboard side at equal distance and one at Forward and Aft side.

(e). The following Fire Extinguishers to be provided at Bridge and Engine

Room;

(i). Dry chemical powder (DCP) Fire extinguishers as per IRS recommendations for Bridge.

(ii). One Nos approved foam type fire extinguisher of 45Ltrs capacity and 2 Nos 9 Ltrs capacity foam type Fire Extinguishers are to be provided for Engine Room.

SECTION “E” PIPING

1. General

(a). Piping shall be designed as per best global standards and classification guidelines. Piping shall be adequately supported.

(b). Routing of pipes shall be done in such a way that, it permits free passage in walking areas, does not affect the performance of the crew. Special attention shall be taken to provide space for maintenance of ship's structure as far as practicable.

(c). A detailed piping single line diagram of each system to be prepared for approval by classification society and owners.

(d). The piping layout shall be such as to provide easy access for maintenance or repairs during service. Wherever piping is taken through W.T. bulkheads, decks or tank tops, approved type deck/bulkhead fittings to be provided to approval of class requirements. Mud boxes, strainers, filters and valves to be provided as required at appropriate places.

2. Piping Material Specification

Material for the piping, valves, flanges and fasteners for piping and other system related material shall be selected as per relevant class requirements.

Treatment of these pipes shall be done in accordance with the relevant class requirements. In general, the materials for various systems shall be as follows:

- Bilge & Ballast -Galvanized Steel
- Fire, Deck wash - Galvanized Steel
- Sea Water - Galvanized Steel
- Fresh Water System- Copper or class approved Plastic (UPVC/CPVC)
- Exhaust - MS Black Steel to IS2062 Gr.B or equivalent.
- Fuel Oil System - MS Black Steel
- Lube Oil system - MS Black Steel
- Engine Cooling water system – MS Black Steel
- Sanitary System - Galvanized Steel

3. FUELSYSTEM

The fuel oil system shall consist of Fuel Oil Service System and Fuel Oil

Transfer System.

Suitable bunkering arrangements shall be provided on the main deck, with suitable connections for filling of the bunker tanks. Class approved sampling system shall be provided at the bunkering station.

3.1. FUEL OIL TRANSFER SYSTEM

Fuel oil shall be transferred from fuel oil storage tank to other storage tank using two (one working and one standby) electrically driven screw type transfer pumps, installed in the engine room.

3.2. FUEL OIL SERVICE SYSTEM

Fuel oil shall be transferred from fuel oil storage tank to each fuel service tank. From the service tanks, overflow lines shall be led to FO Overflow tanks.

Fuel oil shall be transferred from service tanks to main engines/diesel generators with the help of integrated fuel oil service pumps.

4. LUBE OIL SYSTEM

Self-contained lubricating oil systems shall be installed for the main engines, gears and drives. The lube oil system shall be as per OEM recommendations of main and auxiliary machinery.

Closed circuit lubricating oil system engine driven pump shall be provided with standby electric motor driven pump. Automatic changeover to be provided for Lube oil standby pumps with Engine driven pumps with suitable alarm.

Suitable number of lube oil tanks shall be provided by the Builder as hull tanks/loose tanks at appropriate locations based on the recommendations of OEMs of major machinery.

5. DIRTY OIL SYSTEM

Dirty oil in the sludge tank shall be discharged by means of the dirty oil / sludge pump or via direct suction from the shore. The tank shall be provided with a high-level alarm.

6. COOLING SYSTEM

Cooling system shall be provided for each propulsion machinery, D.G. sets, auxiliary machinery, HVAC and hydraulics. In general, the propulsion and auxiliary equipment cooling system are to be based on tube/plate heat exchanger.

A fouling factor of 15% is to be considered for plate/tube type heat

exchangers with back flushing system.

7. COMPRESSED AIR SYSTEM

3 nos. electrically driven starting air compressors and air receiver bottles with capacity and working pressure adequate for the selected diesel engines shall be provided complying to class requirements. Service air requirements shall be met by providing adequate number of reducing stations from the main receiver units.

Service lines shall be provided for sea chest blow down, controls, and various equipment. A main deck service line shall also be provided.

8. EXHAUSTSYSTEMS

The exhaust gas system for each engine/generator shall contain the following equipment:

- i. Exhaust silencer (approx.35dBA and 125Hz-for Main Engines and Diesel Generators)
- ii. Integral exhaust gas spark arrestor

The exhaust system shall be designed such that it provides maximum isolation of machinery vibration and exhaust noise. Removable blanket type insulations are provided for the exhaust gas lines. The entire system shall be resiliently mounted, if required.

9. AIR VENT SYSTEM

Air vents shall be installed for all tanks having filling or suction connections.

Air vents in fuel oil tanks and sludge/oily water/ballast tanks (as applicable) having flammable hydro carbon shall be provided with flame screen. Air Vents for sea chests, sewage tanks shall be provided with goose neck arrangement above main deck.

Diesel Oil bunker stations shall be arranged on both port and starboard side. Connectors for fuel oil filling shall be of cam lock type. For fresh water filling, threaded type connectors shall be provided.

10. SOUNDING SYSTEM

10.1. MANUAL SOUNDING SYSTEM

Sounding provisions shall be provided for all tanks as far as practicable. Manual Sounding system shall be in general of sounding pipe type. Level gauges to be provided for all service tanks, if possible.

10.2. TANKS

(a). Tanks shall be provided to meet the services of the various piping systems. All tanks shall be of mild steel with pipe fittings, suction pipes, air pipes, sounding pipe/level gauges, drain plugs, inspection man holes and shut off valves.

(b). Ship builder shall decide the location of tanks to suit trimming conditions and other construction requirements.

(c). The Main Tanks shall have the following capacities as far as practicable;

- Fuel Oil Storage Tanks: 2 x 25 tonnes (minimum).
- Two daily service fuel oil tanks in engine room for two main engines and for Auxiliary Engines also
- The tanks should be at an elevated position.
- Fresh water tanks for domestic service of about 25 tonnes.
- Foam tank capacity shall be sufficient for 30 minutes continuous operation.
- One expansion tank shall be arranged in common for port and starboard main engines and auxiliary engines.

(d). Fuel Oil service tank shall be provided with facility to drain the oil to bottom storage tank and shall also be provided with test cock. The overflow of service tank to be connected to storage tank.

Other tanks to suit the capacity of machinery shall be installed.

SECTION “F” ELECTRICAL

1. All the electrical equipment and machinery and associated systems on the Tug shall conform to classification rules.
2. **Labelling and Tally Plate**: Anodised Aluminium tally plate shall be fitted on items of electrical equipment and fittings indicating relevant parameters make, serial number, rpm, voltage , current etc.
3. **Tally Plates**: All DBs, NLP, equipments etc shall indicate the nomenclature of the boxes and the details of incoming and outgoing circuits with the capacity of each fuse and MCB there in.
4. **Diagram Plates**: All electrical equipment, switchboards, panels, DBs etc. shall be provided with Anodised Aluminium diagram plates, clearly showing connections with terminal markings on the equipment and they shall be secured on the underside of the cover of the equipment.
5. **Fuses**: Where Fuses are used for circuit over current protection they shall conform to international standards and shall be approved by Classification Society.
6. **Colour scheme of electrical equipment**: Bare metal surfaces of electrical equipment shall be painted with rust proof paint and finishing paint, except those specially treated and rendered corrosion free.
7. **Marking of Bus-bar and Connections**: Bus-bar and connections of major power equipment shall be clearly marked by means of tapes, paints and equipment name.
8. **Caution Board**: All equipment connected to working voltage above 125V AC shall have caution board affixed in a prominent position. The caution board shall be red in colour with white letters. Framed notice plates demonstrating method of Classification Society Rule Aid for electric shock shall be filled in the proximity of the switch board.

9. **Alternators:** Two in number DAs of each 140 KVA capacity with each capable of taking 100% of maximum load with one remaining standby, conforming to classification rules to meet electrical loads satisfactorily under various operating conditions. Final rating shall be as per electrical load calculations. The DAs shall be self-regulated & self-excited. An analysis of electric power (load chart) during operating conditions for the main alternator are to be submitted to the buyer.

Power Rating	140 kVA(approx.)
Voltage	3 Phase 415 V, 50 Hz, 3 wire
Insulation	Class-F/F
IP	as per class rule
AVR	Solid state design or as per maker standard
Excitation	Self-excited, self-regulating type with brushless excitation system
Type of cooling	Engine shall be FW/SW cooled and Alternator shall be air cooled or water cooled as per OEM Standard.

10. Voltage and frequency tolerances shall be within the limits specified by the Class rules. Generator space heaters shall also be provided for each generator. The design of the electric plant including generators, motors and controllers shall be coordinated to ensure that the voltage dip, when starting the motor with highest inrush current shall not exceed 15% of the rated voltage.
11. Alternators are not normally required to run in parallel continuously. It shall, however, be possible to parallel the alternators for the purpose of changing over of load from one alternator to the other without interruption.
12. **Harbour Alternator:** One 65 KVA Diesel driven, harbour alternator shall also be provided to meet essential services when the Tug is alongside jetty and when main alternators are shut down. Final rating shall be as per electrical load calculations.

13. **Power supply requirements:** Power supplies shall be provided as follows:-
(a) 415V AC, 3 Phase, 50Hz, 3 wire, ungrounded system
(b) 230V AC, 50Hz, 3Phase, 3 wire, ungrounded system
(c) 230V AC, 50Hz, 1Phase, 3 wire, grounded system for domestic loads
(d) 24V DC
14. Emergency Lighting / Emergency lighting fittings etc., supplied from 24 Volts DC distribution board shall be fitted to provide illumination to access ways, passage ways, in each compartment and as approved by class rules.
15. **Transformers:**
(a) 2 no. of step down transformers with 3 phase, 415V, 50 Hz, 3 wire Delta connected primary to 3 phase, 230V, 50Hz, 4 wire Star connected secondary with grounded star point to cater for Domestic/ portable and COTS equipment. One transformer acts as redundant.
(b) 2 no. of step down transformers with 3 phase, 415V, 50Hz, 3 wire Delta connected primary to 3 phase, 230, 50Hz, 3 wire Delta connected secondary to cater lighting, communication equipment, and navigational loads. One transformer acts as redundant.
(c) Transformers shall be Drip-proof, air cooled, suitable for marine use, continuous rating and 'F' class insulation.
- Shore Supply Arrangements:**
16. A water tight Shore Supply Connection Box of 415V, 3 Phase, 50 HZ will be fitted on weather deck at an appropriate position to meet harbour loads. Shore supply cable of 100 meters length shall be supplied and suitable stowage arrangement shall be provided for the shore supply cable. Shore supply box must be conforming to IP 57.
17. Electrical interlocking of main generator breakers and shore connection breaker will be provided in such a way that the shore connection breaker cannot be switched "ON" if any one of the main generator breakers is "ON" and vice versa.
18. **24V DC Battery Charger & Distribution Board:** A transformer rectifier shall be provided as per the requirement. The charging

facilities to include boost and trickle charge while floating. The rectifier input shall be either 230V single phase supply or 415V, 3 phase supply based on the load requirements. The scheme for transformer rectifier shall be clearly indicated. The rectifier shall have growth margin of 20% for future power requirements. During normal condition it will be used for both charging batteries and feeding power supply to emergency consumers. Upon failure of main source of electric power, accumulator battery shall feed power supply to emergency consumers through emergency switchboard.

19. **Batteries:**

Emergency source of electrical power is an accumulator 24V battery. Battery backup/ UPS will be provided for critical navigation, communication, propulsion/ power generation controls and lighting in case of total power failure.

(a) One set of storage batteries shall be installed in case of electric started engines. The batteries shall be provided in well ventilated spaces with hard wood boxes.

(b) One set of storage batteries for navigation & communication equipment shall be installed either in a battery room or inside dedicated battery boxes placed at suitable location.

20. All essential consumers are required to be provisioned with battery back-up/ alternate supplies in accordance with Class requirement for safety of the tug.

21. **Main Switch Board:**

(a) The MSB shall be of enclosed sheet-steel cubicle suitable for marine duty, having self-supporting frame work. The construction of the panel board shall be of IP 23. The panel boards shall be manufactured from 14 SWG primed steel plate.

(b) A Main Switch Board (with bus bar coupler between each generator section), switch gear and distribution boards conforming to Classification Society rules shall be provided. The design of the switchboard shall be such that installation cabling and maintenance including removal and replacement of components shall be provided from front. The main switchboard shall be certified by Classification Society and Classification Society test certificate for main switchboard shall be provided by the Seller.

- (c) Sufficient number of distribution boards for 415V 3 phase 50Hz & 230V 3 phase 50Hz, 230V single phase and 24V DC shall be installed at appropriate positions as per requirement.
- (d) A supply breaker with each generator and shore supply connection box with necessary interlocking arrangement shall be provided. Electrical interlock shall also be provided between the generators and shore supply so that only one generator or shore supply can feed the bus-bar at a time.
- (e) One 230V feeder panel with Voltmeter, Ammeter and indicating lamps will be provided.
- (f) Following protections, controls, instrumentation & indications for DG set shall be provided in the main switchboard:-
 - (i) **Protections:** Over load protections, under voltage protection, short circuit protection and reverse power protection, bus tie breaker.
 - (ii) **Controls:**
 - (aa) **Main Generators:** Governor control switch, synchronizing switch, MCCBs of suitable rating for manual switching and Electrical/ manual tripping. Provision for manual and semi auto synchronizing and paralleling of generators shall be arranged.
 - (ab) **Shore Supply:** Manually operated MCCB with Electrical/ Manual tripping feeders. Each feeder will be controlled by a MCB of suitable rating.
 - (iii) **Instrumentation:** Main generators/ Shore supply- Ammeter with phase selector switches, current transformer, Voltmeter with phase selector switches, frequency meters, KW meters (only for main generators) and Synchro scope (only for main generators).
 - (iv) **Indication:** "Generator running" lamps, "Bus alive" lamps, "Shore supply On" lamps, "Synchronizing" lamps, "Earth fault" lamps.
- (g) Each section of the switch board shall be provided with anodized wiring diagram at a suitable location (preferably inside the door,

properly fixed) to facilitate fault diagnosis/ rectification and flame retardant tallies for control switches, instruments, control indications, fuses etc.,.

- (h) Non-conducting handrails in front and oil resistant rubber insulating mats with non-skid surfaces in front and back of the switch board shall be provided.
- (i) All bus bars shall have sufficient current carrying capacity for continuous operation and provisions shall be made for withstanding mechanical strains. Bus bar shall be of hard drawn high conductivity copper.
- (j) Fire retardant insulation materials shall be used for insulating purposes. A portable CO₂ extinguisher is to be provided in vicinity of the Switch Board.
- (k) All cables entering the switch board shall be fitted with approved glands.
- (l) Class Test Certificates shall be submitted for Main Switch board.
- (m) No live parts shall be less than 12" from deck

22. **Distribution Boards (DBs):**

- (a) Main supply shall be derived from the main switch board through feeder MCCBs/MCBs. Further distribution of the supply shall be arranged through distribution boards located at load centers. Distribution Boards/ Navigation light panel shall be of watertight construction and of minimum dimensions.
- (b) All DBs of watertight construction with MCCB/MCB protection shall have minimum spare feeder ways as mentioned below:-
 - (i) DBs of 6 ways and below - 1 feeder
 - (ii) DBs of 8 ways and above - 20% of total feeders with minimum of 2 feeders

23. **Starters/Control Panel:**

- (a) Direct on-line motor starters shall be provided for all services below 5HP including portable pumps and Star-delta Starters for

motors 5HP and above incorporating under-voltage, over-current, short circuit, single phase protection and connection for thermistors in the motor.

(b) The starters for Engine room ventilation fans shall be provided with a remote stop push button. These push buttons will be fitted outside the engine room near the entrance to the space. The push button shall be recessed type to avoid inadvertent tripping of the starter. Contactors and overload relays as per approved list of IRS will be used.

(c) Starters located outside the engine room will additionally be provided with a main switch/ isolator with door interlock. Starters for motors will generally be grouped as follows:-

- (i) Group I - Ventilation fans
- (ii) Group II - Engine room auxiliaries
- (iii) Group III - Other services

(a) Remote "Stopping" of the group(s) during "Emergency" will be provided at:-

- (i) Wheel house
- (ii) Engine room exit

(b) Construction shall be such that all installation, main cabling & maintenance work including removal & replacement of components can be performed from the front side of the control panel.

(c) Emergency stop system complying with Classification Rules shall be provided

Motors:

24. All motors shall conform to classification rules suitable for marine use. Each motor will have thermistor / corresponding relay in starter panel.

25. Steering motors, panel and steering potentiometer should be located in the same compartment to facilitate physical sighting of motors during running/maintenance.

Cables:

- 26. Marine grade XLPE or EPR insulated cables with Low Smoke Zero Halogen (LSZH) properties conforming Class requirements, shall be used. All power cables shall be approved by the classification society.
- 27. Each and every cable shall be affixed with brass identification tallies at each end before entering equipment. The tally shall indicate circuit code number as shown on relevant drawings.
- 28. Cables under floor in engine rooms will be run through metallic rigid/flexible conduits. All wiring will be in accordance with IRS class requirement.

29. **Light And Signal Equipment:**

- (a) **Main Navigation light indication and control panel:** A galvanic isolated 24V power supply system to be provided for navigation and signal lights. The primary power supply system to be from 230V main system and with changeover to emergency supply in case of failure in main system. Navigation lights shall be provided as per Class Rules/COLREGs
 - (b) **Air Whistle:** An air whistle shall be positioned on the superstructure, with its operating controls in the bridge.
 - (c) **Bell:** One bell to be provided according to Regulatory requirements.
 - (d) **Search Lights:** Two 1000 watt search light without shutter to be fitted on wheel house top where maximum search area can be covered and shall be remotely controlled from wheel house.
- 30. **Flood Lights:** 4 Nos. 250W, LED flood lights to be provided for night working one each at tow hook/capstan on aft and windlass at ford of main deck. One additional power point will be provided at tow hook on aft of the main deck for flood lights.
 - 31. **24V Emergency Lights:** Adequate no. of 24V DC operated Emergency light fittings shall be fitted to provide illumination to access ways and passageways. The emergency lighting system shall be as per class rules. In the event of failure of the ship's AC normal lighting 24V DC emergency lights to be switched on automatically.

32. **Cabin Fans:** Cabin fans of suitable sizes operating on 230 V AC to be provided in accommodation spaces, offices and manned stores/spaces.
33. **Sockets:**
- (a) Suitable number of sockets outlets, 230V, 15 A to be installed in engine room and workshop.
 - (b) Non-water tight 230V, 5A sockets to be installed in cabins, public spaces, lockers etc.
 - (c) Two sockets shall be fitted in each cabin.
 - (d) Two off three phase sockets of suitable rating for welding machines are to be provided at suitable locations as agreed between the Buyer and the Builder.
34. **Bridge Console:** A console shall be provided in the wheel house to accommodate all necessary controls for Propulsion units, auxiliary machinery, Navigation Lights Control and indication panel, SP telephone, VHF, Echo Sounder, Speed Log, Horn Control, etc.,
35. **Clear View Screen (CVS):** Clear View Screen conforming to classification rules shall be provided.
36. **Navigation Equipment:** Navigation equipment is to comply with the requirements for Coastal Vessel rules as per DGS Order 01 of 2014 or its latest amendment as applicable at the time of vessel construction.
- (a) **Radar:** One X-band ARPA radar, 12 kW, 4 ft scanner and 23.1” LCD display according to manufacturer’s standard.
 - (b) **GPS:** One GPS shall be installed
 - (c) **Magnetic compass:**
 - One Magnetic Compass with Azimuth Mirror for terrestrial navigation to be installed.
 - One spare compass in box to be delivered.
 - Certificates and deviation schedule to be delivered.
 - (d) **Echo Sounder:**
 - Navigational echo sounder to be provided.
 - (e) **Other Nautical Equipment:**
 - One Automatic identification system (class A) to be installed
 - One set of anemometer to be installed

37. **Communication Equipment:** External and internal communication equipment to comply with the requirements for coastal vessel as per DGS order 01 of 2014 or its latest amendment as applicable at the time of vessel construction.
- (a) Two VHF/DSC Radio Telephone with built in VHF/DSC Watch Receiver
 - (b) Three no. of Two-Way VHF hand held Radio
38. **Calling, Command And Telephone System:**
Separate systems shall be provided for talk-back and loudhailer system.
- Telephone System/PA System**
- (a) Telephones to be provided in Wheel house, MSB room, Cabins etc. Number of telephones and their locations to be finalized during contract stage.
 - (b) Loudspeakers for PA function to be mounted in public area such as corridors, mess/cabins, etc.
39. **Automation Systems For Machinery**
- (a) **Alarm System:** Comprehensive alarm with monitoring system shall be provided as per class and OEM's requirement for major machinery including Main engines, auxiliary engines and critical ship systems etc.
 - (b) **Engine & Azimuth Thruster Controls:** Remote operation shall be so designed that the propulsion plant can be operated from the Bridge. Instrumentation/alarms shall be as per class rules/maker standard. Engine controls shall be arranged in the Bridge as follows:
 - E/R control with main engine maneuvering devices, auxiliary device, instruments etc.
 - Generator/auxiliary engine alarm and control panels.
 - Intercom stationEmergency control shall be provided, with which the engines can be controlled mechanically in case the remote-control system failure. UPS shall be provided for Engine Control System.

- (c) **Diesel Generator Controls:** The generator engine shall be speed controlled remotely by governor control switch on main switch board and also controlled at engine side. Instrumentation/alarms as per class rules and OEM standards shall be provided. Control power supply for the DGs shall be taken from 24V power supply provided on board. Local engine control panel with necessary alarms, indication lamps and control switches shall be provided.
- (d) **Fire Detection System:** In general, fire detection system shall be provided adhering to relevant Class/Statutory Regulations. This shall consist of an addressable fire & smoke detection system in all compartments, including the engine room and all stairways, corridors, escape routes within accommodation spaces, etc. The detectors shall be of smoke/heat or thermal type dependent on location of installation.

SOURCES OF SUPPLY FOR EQUIPMENT FOR 60T BP TUG

HULL EQUIPMENT

Sl. No	Equipment/Material	Probable List of Vendors
1	Anchor capstan, Mooring Capstan, Anchor Windlass	(a). M/s. GRSE Ltd, Kolkatta (b). M/s. Geeta Engineering Works (c). M/s. Hemant Engg Works (d). M/s. H & H Precision Pvt Ltd.
2	WT/GT Doors WT Hatches	(a). M/s. Godrej & Boyce Mfg Ltd. (b). M/s. Larsen & Tubro, Talegaon. (c). M/s. SGS Marine Habitability Pvt Ltd.
3	Anchor	(a). M/s. Chakraborty Traders & Engineers (b). M/s. Govardhan Das PA (c). M/s. Indian Chain Pvt Ltd
4	Anchor Chain cables and accessories like Swivel adapter, adapter piece etc	(a). M/s. Indian Chain Pvt Ltd (b). M/s. Atlantic Links (c). M/s. Butt Weld Industries.
5	Sacrificial Anodes	(a). M/s. Sargam Metals Pvt Ltd. (b). M/s. Cathodic Control Co Pvt Ltd

ENGINEERING

Sl. No	Equipment/Material	Probable List of Vendors
1	Main Diesel Engines	(a). M/s ABC Engines (b). M/s. Bergen
2	Steerable rudder propeller along with gearbox/ clutches	(a). M/s Schotell
3	External fire fighting	(a). M/s FFS (b). M/s Jason (c). M/s Marsis
4	Diesel Generator	(a). M/s Greaves Cotton (b). M/s Koel
5	Harbour Diesel Generator	(a). M/s Greaves Cotton (b). M/s Koel
6	Centrifugal pumps	(a). M/s Desmi (b). M/s SPX
7	Positive displacement	(a). M/s Desmi (b). M/s SPX

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	pumps	
8	Portable pumps	(a). M/s Mody pump (b). M/s WPIL (c). M/s Harison pump
9	Air compressors	(a). M/s Elgi sauer Compressors (b). M/s Ingersol rand

ELECTRICAL

Sl.No	Equipment/Material	Probable List of Vendors
1	Alternators	(a). M/s. Crompton Greaves. (b). M/s. Elmot, Hyderabad
2	Main Switch Board	(a). M/s. L&T, Mumbai
3	Motors	(a). M/s. Bharat Bijlee, Mumbai
4	Starters/ Control Panels	(a). M/s. L&T, Mumbai
5	Transformers	(a). M/s. Static Transformers, Indore (b). M/s. Auto Electronic control, Mumbai (c). M/s Ankur Enterprises
6	Transformer Rectifiers	(a). M/s. Marine Electricals, Mumbai (b). M/s. Auto Electronic control, Mumbai (c). M/s. Static Transformers, Indore (d). M/s. Wave Electronic, Kochi
7	Distribution Panel/ Navigation Light Panel/ Distribution Boxes/ Test Panel	(a). M/s. Marine Electricals, Mumbai (b). M/s. Auto Electronic control, Mumbai (c). M/s. Arvin, Mumbai (d). M/s. Wave Electronic, Kochi (e). M/s. Ray Enterprises, Ambala cant (f) M/s. Ankur Enterprises
8	Light Fittings	(a). M/s. Arvin, Mumbai (b). M/s. Manish Industries, Kolkatta (c). M/s. Ray Enterprises, Ambala cantt (d). M/s. ISSAC, Kolkatta (e). M/s. Sipani Energy Ltd (f). M/s. Zealtech Ltd. (g). M/s MCGEOCH Marine Electricals Pvt. Ltd.
9	Fans/Blowers	(a). M/s. Almonard Ltd. (b). M/s. Orient
10	Plug & Socket	(a). M/s. Arvin, Mumbai (b). M/s. Manish Industries, Kolkatta

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		(c). M/s. Ray Enterprises, Ambala cantt (d). M/s. ISSAC, Kolkatta
11	Shore Supply Box	(a). M/s. Marine Electricals, Mumbai (b). M/s. Wave Electronic, Kochi (c). M/s. Neo Power Electronics & Projects (d) Ltd, Mumbai.
12	Cables	All OEMs who have works approval certificate from any IACS member
13	Batteries	(a). M/s. Amara Batteries, Chennai (b) M/s. Exide, Kolkatta
14	Navigation Lights	(a). M/s. Arvin, Mumbai (b). M/s. Manish Industries, Kolkatta (c). M/s. Ray Enterprises, Ambala cantt (d). M/s. ISSAC, Kolkatta
15	Marinised Electric Horn (Class Approved)	Can be sourced from any Reputed Vendor.